

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“If PCs had two form factors, and mobile computing had tens of form factors, the IoT will have millions of form factors”

Guru Ganesan, managing director of ARM India, speaks to Janani Gopalakrishnan Vikram about the ‘diversity’ in the Internet of Things (IoT), the kind of processors needed for it to grow, and how these processors foster innovation in the IoT ecosystem

What do you think is the current scope of the IoT?

If PCs had two form factors—the desktop and laptop, and mobile computing had tens of form factors—from candy bar, flip, slide and touchscreen cell phones, to smartphones, tablets and phablets, the IoT will have millions of form factors. This will bring about immense choice and diversity. There will be millions of different devices and apps.

The IoT will be an enabler of new products, services and consumer behaviour. It will be driven by whoever has ideas.

For consumers, the IoT is about personal use, convenience and health, also known as the ‘the quantified self,’ a movement to incorporate technology into data acquisition on aspects of a person’s daily life in terms of inputs (e.g., food consumed or the air quality), states (like moods, state of arousal, blood oxygen levels), and performance (mental and physical). Such self-monitoring and self-sensing combines wearable sensors (EEG, ECG, video, etc) and wearable computing as exemplified by the Nike fuel band.

What kind of processors do IoT products require?

To enable the IoT, the electronics industry needs to deliver processor

technology to power everything from tiny ultra-low-power sensors that intelligently gather information, and the gateways and networking equipment that collate and transmit the data, to the cloud-based servers that store and analyse the data for retrieval, often via smartphones or mobile devices. Each of these stages will require high-performance but low-power processor technology.

ARM is uniquely able to provide processors for each of these stages, from sensors through mobile devices to servers.

How do you spur diversity and innovation in the IoT ecosystem?

ARM’s business model is based on licensing core technology to different partners, allowing them to differentiate and add value. Having multiple suppliers for each form factor ensures competition, and competition spurs innovation and differentiation, which exemplifies the fact that diversity is what the IoT is all about.

Besides the ability to address different markets, to explore new ones, and the ability to scale from small to large, the ecosystem for the IoT will have to be more diverse. Enabling ecosystems that drive innovation is a critical part of what ARM does with its partnership-based business model.

While ARM’s partners sold 8.7 billion chips in 2012, the total market for embedded control was 17.2 billion chips. By 2020, the prediction is that the market will have grown to many billions of chips driven mainly by IoT devices. Many of those embedded controllers are in products that will become part of the IoT in the future.

Can you tell us about some classic IoT products that use ARM processors or IP?

AlertMe is an affordable platform that enables consumers to monitor, control and intelligently automate their homes. It uses a multi-protocol home gateway with ARM9 processor-based technology that communicates with home devices incorporating ARM Cortex-M3 processors.

ArmAssist is a rehabilitation device worn on a patient’s arm that measures the movement of the shoulder and elbow through an orthotic or orthopaedic brace. It is built on a 32-bit Flash microcontroller from STMicroelectronics’ STM32 family, based on the ARM Cortex-M3 processor.

The Talking Book is a low-cost, handheld audio computer designed specifically for the learning needs of populations that cannot read, in the poorest regions in the world. It utilises an ARM Cortex-M0 processor.